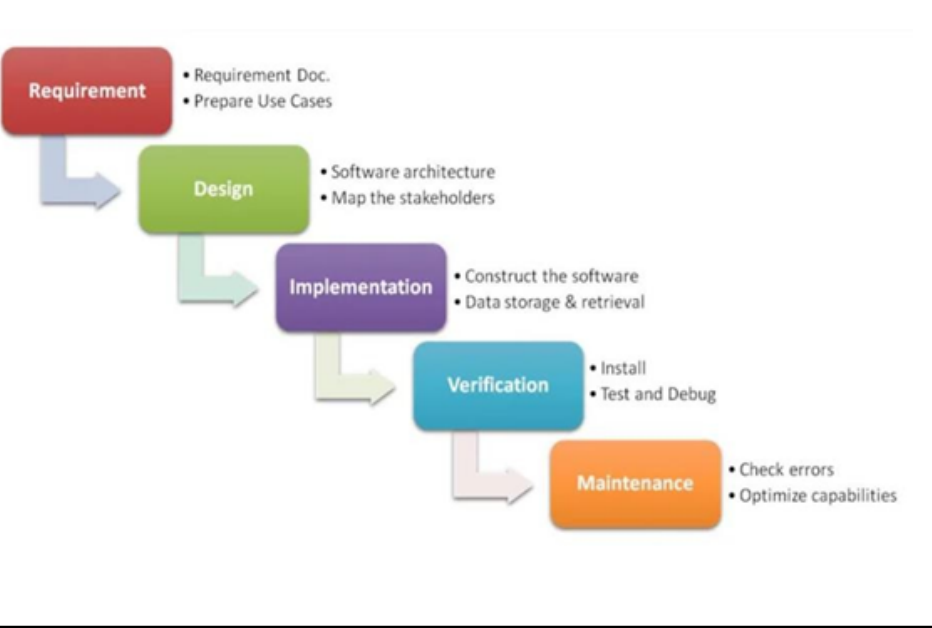
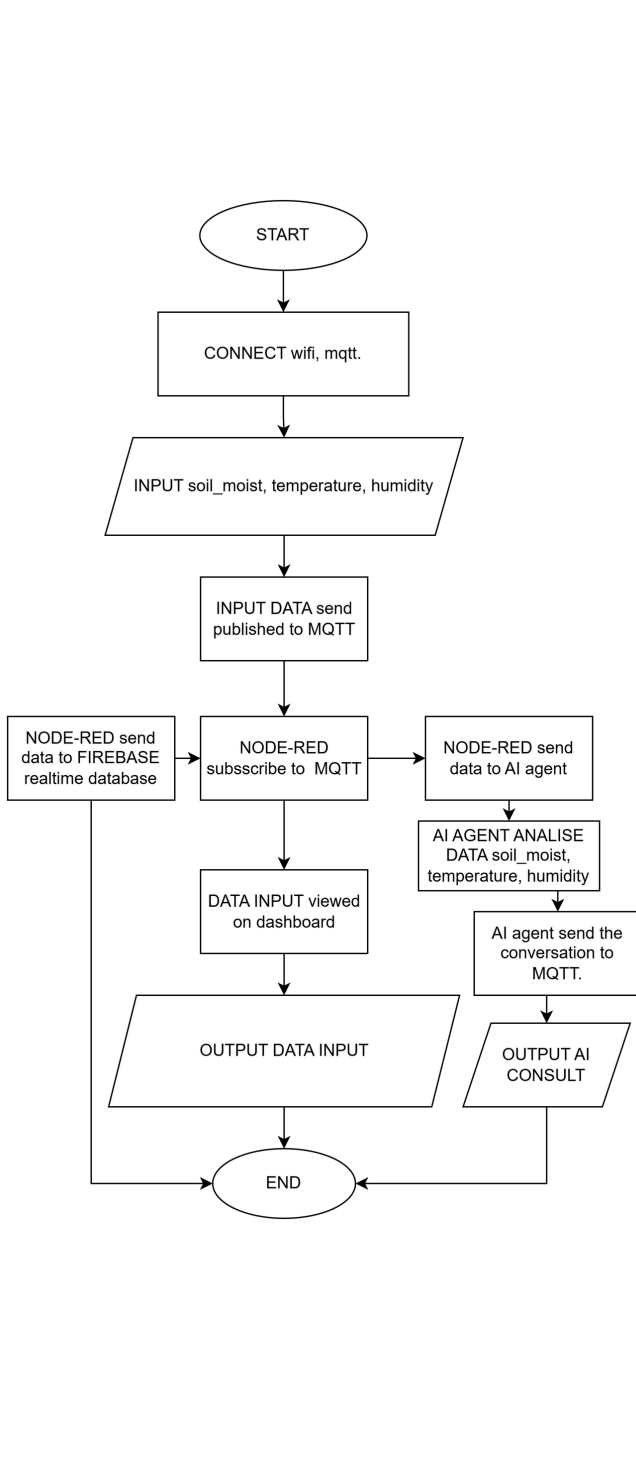
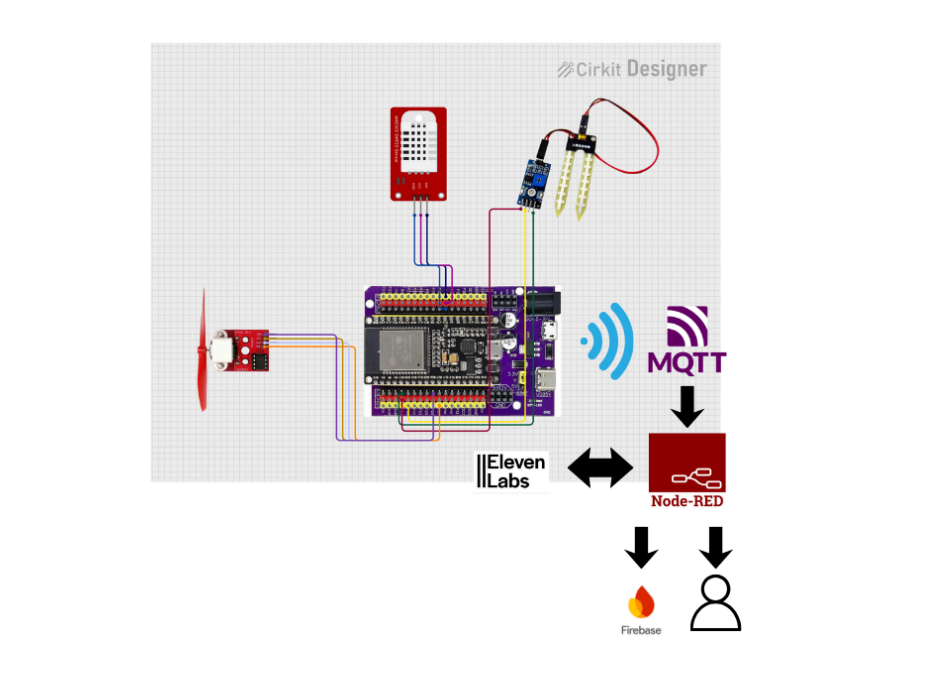
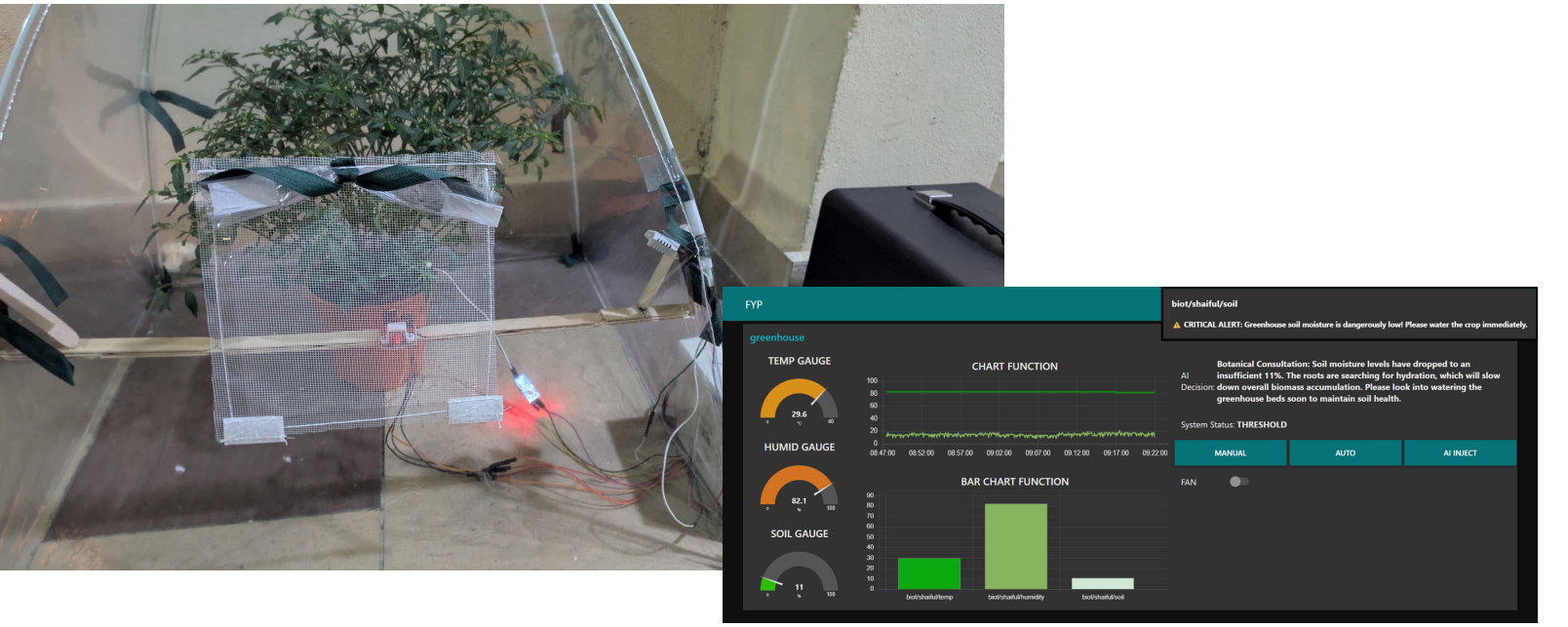


INTRODUCTION		METHODOLOGY	FLOWCHART
Traditional greenhouse monitoring relies on manual checks, which can lead to delayed responses to plant stress. While IoT integrations offer real-time tracking, raw data is often difficult for non-technical users to translate into immediate action. This study proposes an IoT-enabled smart greenhouse that utilizes cloud telemetry for robust data logging, paired with an AI-driven botanical consultation engine. This system bridges the gap between raw hardware metrics and actionable human understanding by providing instant, natural-language crop health advice and automated alerts.			
PROBLEM STATEMENT		LAYOUT	
- Lack of Real-Time Monitoring. - Data Interpretation Barriers. - Hardware Instability in Edge Systems..			
PROJECT OBJECTIVE		RESULT AND DASHBOARD	
- To design an IoT based greenhouse system that monitors and collects the environment data such as temperature, humidity, soil moisture using sensor. - To implement cloud based ai agent capable of analyzing real time sensor data and making consultations. - To demonstrate the effectiveness of combining IoT and AI technologies in achieving precision agriculture and sustainable greenhouse management.			
SCOPE	LIMITATION		
- Hardware Integration - Software & Cloud Architecture - Data Analytics & AI	- No Autonomous Irrigation - API Dependency - Small-Scale Deployment		
FUTURE RECOMMENDATIONS			
Isolated Autonomous Irrigation: Reintroduce physical water actuators by utilizing optocouplers and decoupled power supplies. This ensures automated crop hydration while strictly isolating the inductive load to prevent electromagnetic interference (EMI) with the edge sensors. Advanced API Orchestration: Scale the cloud architecture by migrating the data routing layer to a dedicated workflow platform like n8n. This enables robust error handling, complex webhook management for external LLMs, and simultaneous multi-database logging. Predictive Machine Learning: Utilize the permanent historical telemetry datasets retained within the Firebase Realtime Database to train a time-series forecasting model. This will shift the system's core logic from reactive threshold alerts to preventative, predictive crop management.			
REFERENCE		CONCLUSION	
- Smart Agriculture: IoT-based Greenhouse Monitoring System Simo et al. (2022) - IoT Based Smart Greenhouse and Smart Farming Mishra et al. (2023) - IoT-enabled Greenhouse Systems: Optimizing Plant Growth and Efficiency Manoharan et al. (2024) - Transforming Data Annotation with AI Agents: A Review of Architectures, Reasoning, Applications, and Impact Karim et al. (2025) - An agent-based service architecture for smart greenhouses Pardo et al. (2026)		This project successfully establishes a highly stable, cloud-connected greenhouse monitoring platform. By strategically pivoting away from unstable physical water actuators toward robust Firebase data logging and intelligent Node-RED threshold notifications, the system ensures continuous uptime. Furthermore, the integration of an AI diagnostic engine successfully transforms raw environmental telemetry into actionable, expert-level agricultural advice directly on the user dashboard.	